

Effective defensive strategies against Real Madrid

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Soccermatics Pro Course

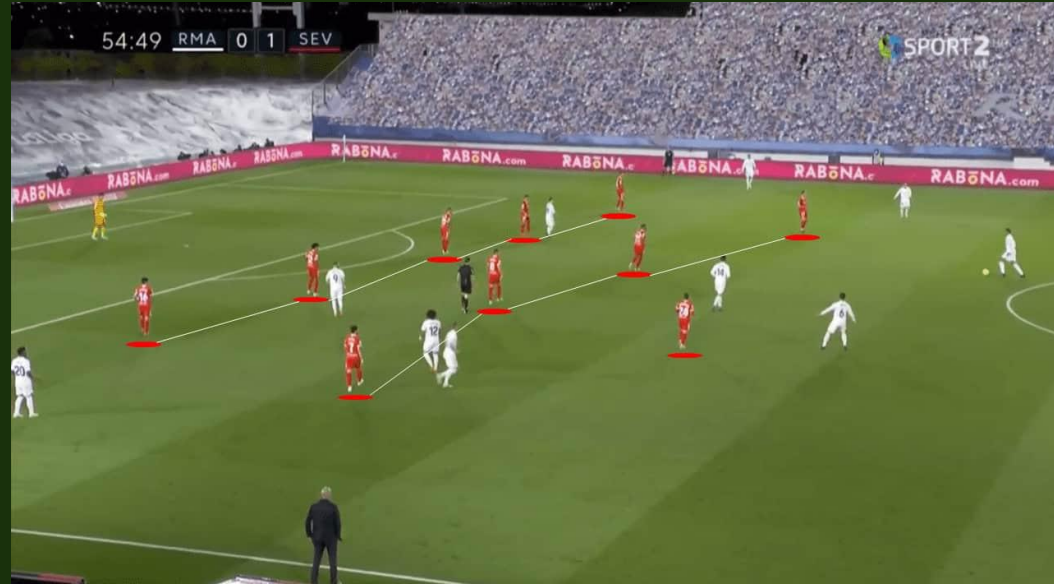
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Problem Description

It is no secret that Real Madrid can field a powerful, fluid offensive capable of scoring multiple goals for any match. They accomplish this by utilizing skilled attacking players who can drag defenders out of position to create space, or create 1v1 isolations in the attacking third.

To counter these abilities, conventional wisdom suggests defending in a tight, low block to prevent exploitable spaces in the defending third, and continuous pressing to avoid 1v1 matchups.

Can analyzing match data give insights into which defensive strategies are most effective against Real Madrid's offense? This study uses SkillCorner tracking and dynamic event data from the 2023/24 and 2024/25 Real Madrid seasons to measure relationships between defending team tactics and Real Madrid offensive metrics.

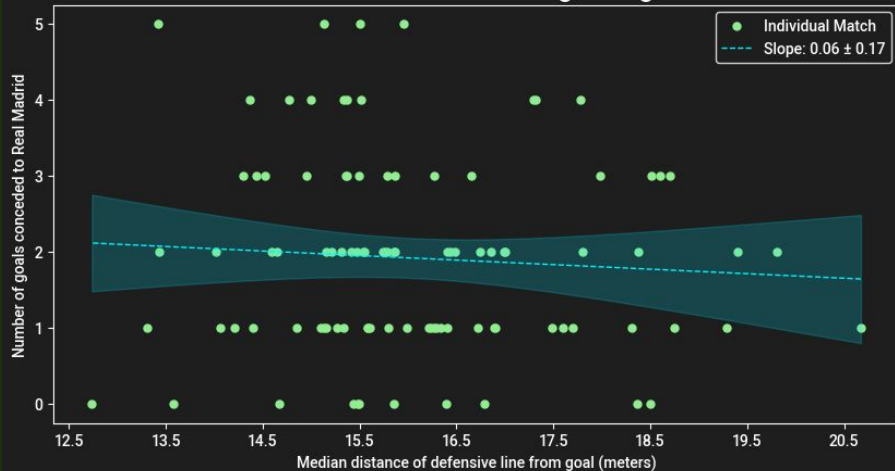


Defensive metric analysis

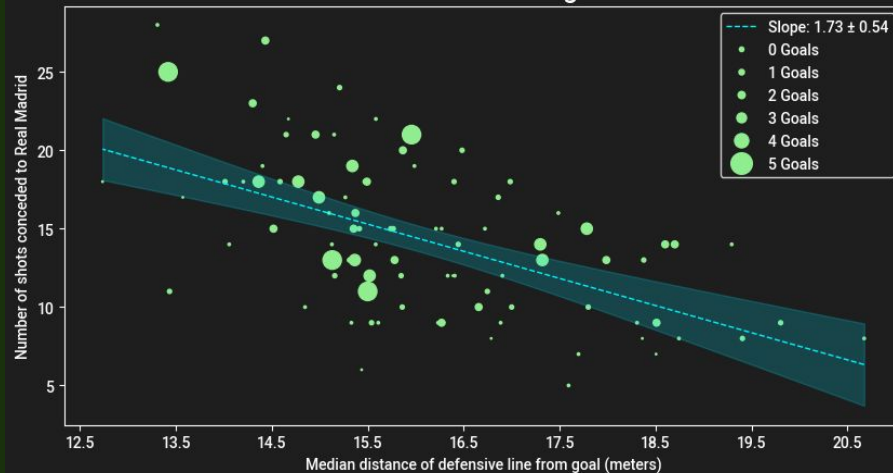
Defensive Line Height

As the defensive line height increases, the number of shots taken by Real Madrid decreases between 1.19 to 2.27 (with 95% confidence) fewer shots for every meter further away from their own goal. Similarly, the xG of those shots also decreases proportionally. However the effect on goals is essentially non-existent, suggesting that Real Madrid is a team that will still get their goals regardless of how further up the pitch the defense pushes them.

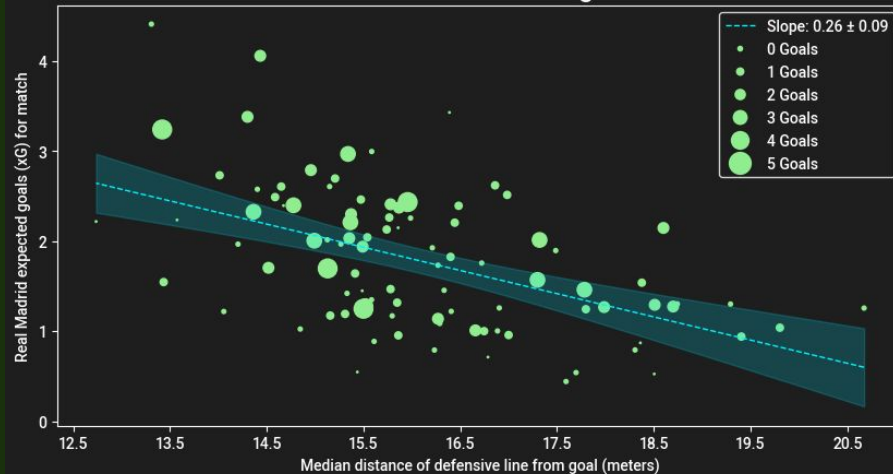
Median defensive line height vs goals



Median defensive line height vs shots

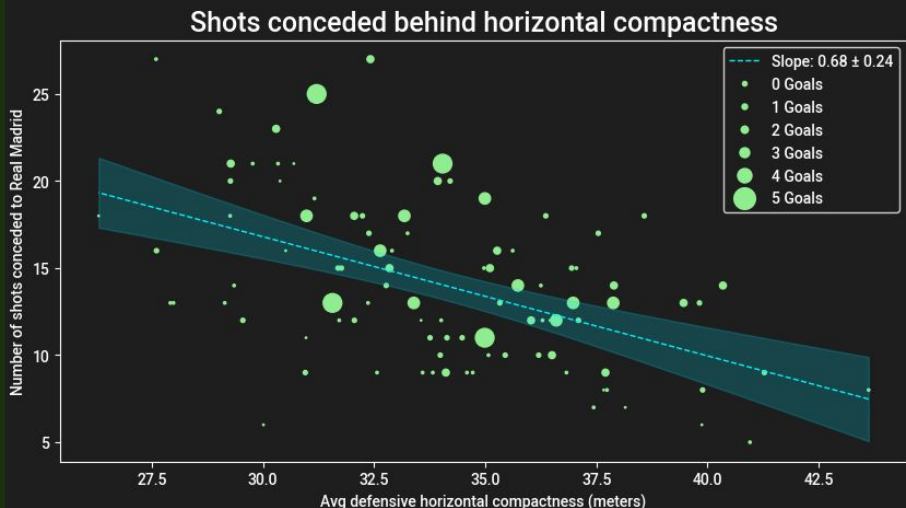
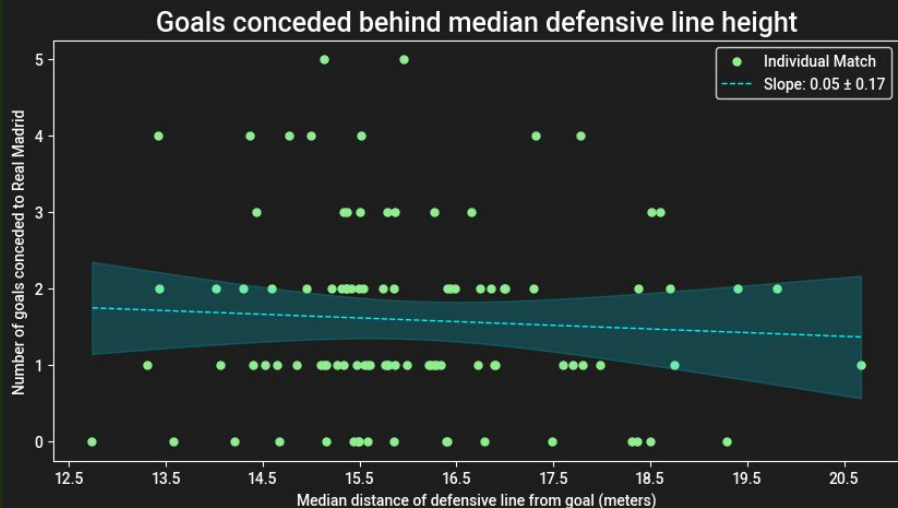
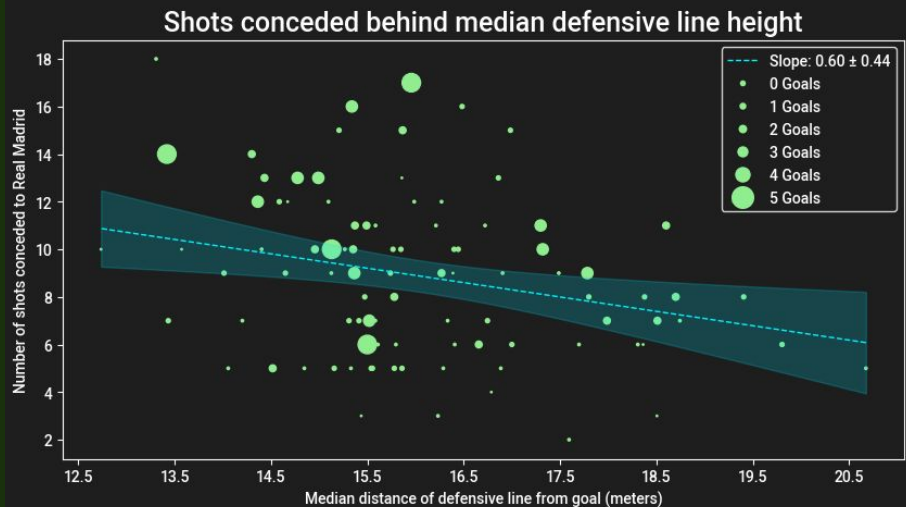


Median defensive line height vs xG



Attacks behind defense

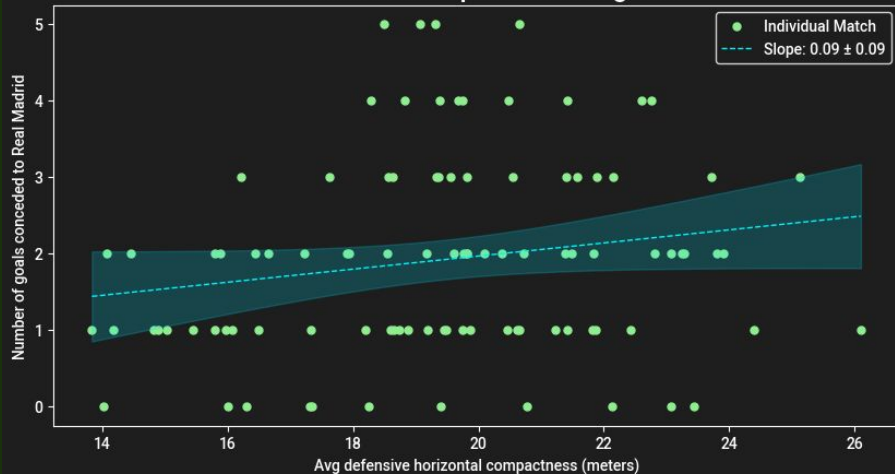
Maintaining a high defense runs the risk of leaving space open behind the defense. These graphs look at the number of shots and goals conceded behind the median defensive line for that match. There are shots that do originate from behind the defensive line, but the trend is less than the overall number of shots. The slope for goals is effectively flat, suggesting that goals can expect to happen behind the defensive line and no specific line height has any effect on goals scored.



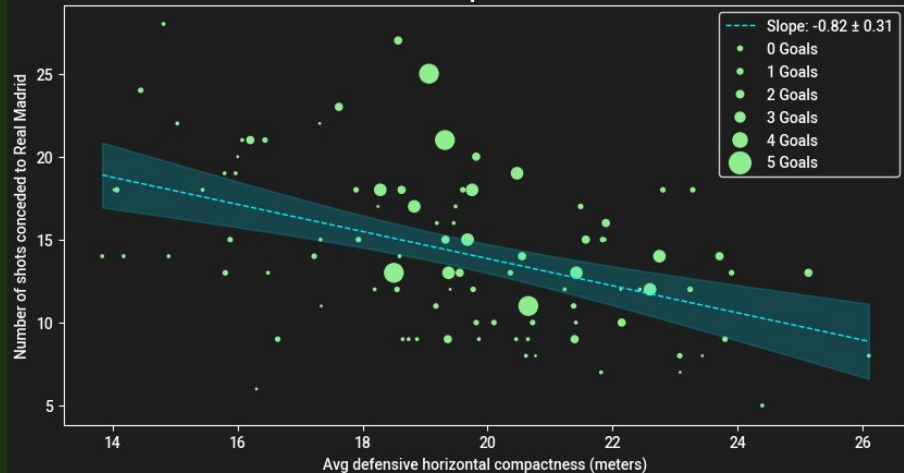
Horizontal Compactness

The average horizontal spread between defensive players, as measured by the last 8 defensive (excluding GK) players closest to their own goal, shows that tighter horizontal compactness when defending actually resulted in *more* shots with higher xG attempted. The same compactness trended towards fewer goals, however the trend is weak with the 95% confidence interval hovering around zero.

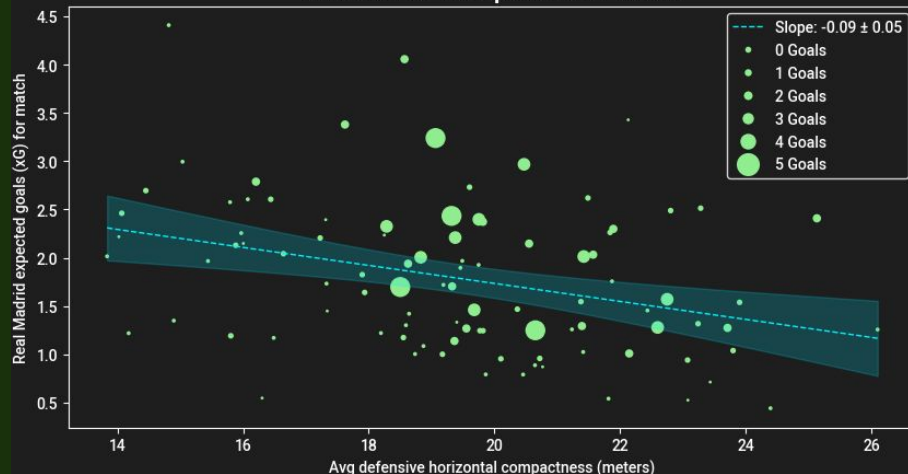
Horizontal compactness vs goals



Horizontal compactness vs shots

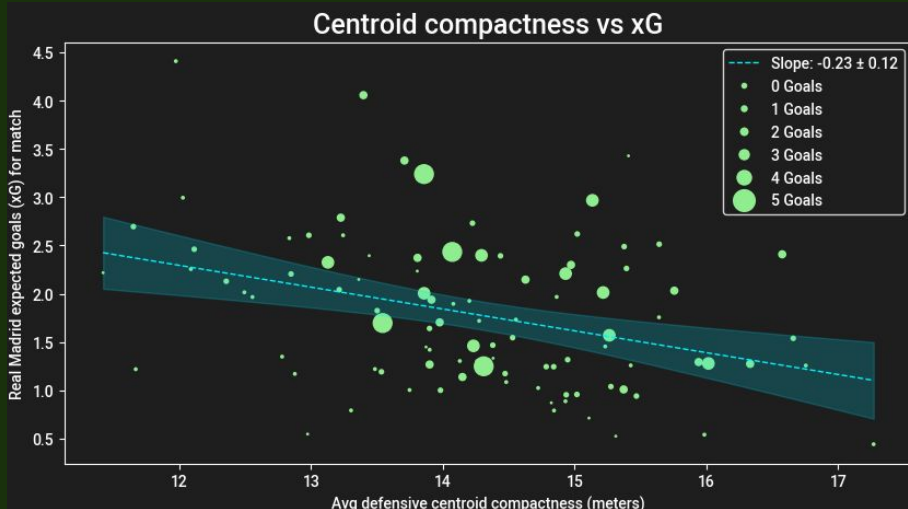
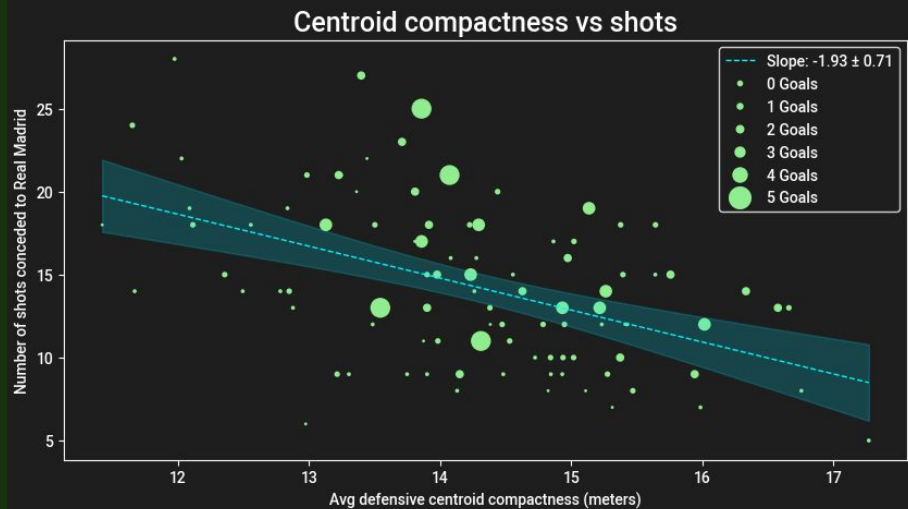
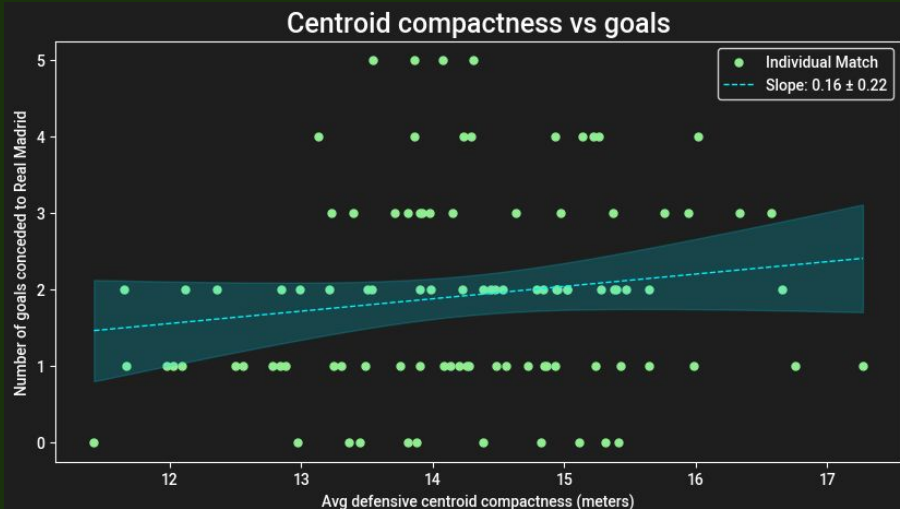


Horizontal compactness vs xG



Centroid Compactness

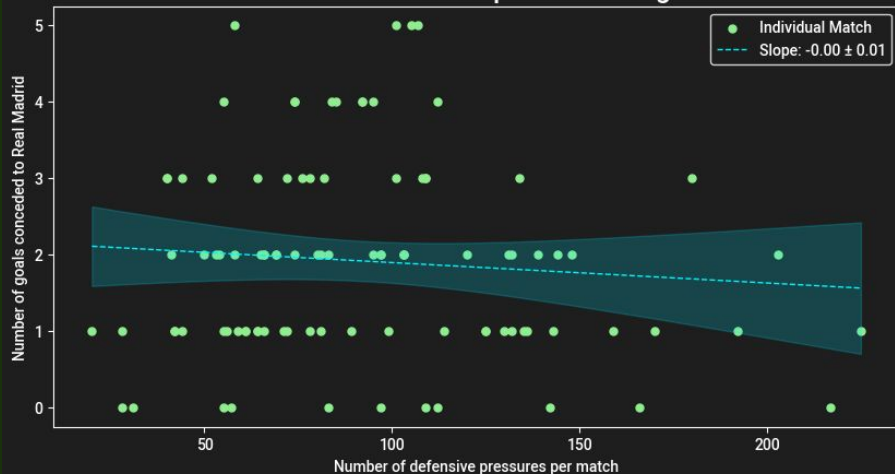
The average compactness over a match, as measured by the average distance of every defensive player from the center point (centroid) between all players, shows similar divergent results. A tighter compactness when defending against Real Madrid leads to *more* shots with higher xG attempted. However the same compactness suggests fewer goals, but only 0.16 fewer goals for every meter of tighter compactness.



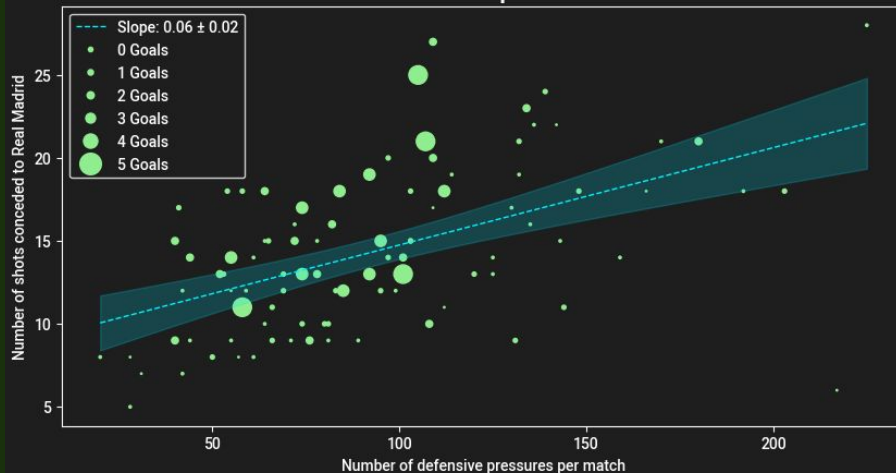
Defensive Pressure

The number of defensive pressures initiated by the defense had the most surprising results. When more pressures are applied, Real Madrid is able to take more shots with higher overall xG. However the number of goals scored is essentially unaffected by defensive pressures, making them largely unnecessary for the extra shots they allow. This can also explain the same trend seen when compactness loosens up if the “breaking away” from compactness is being used for pressures.

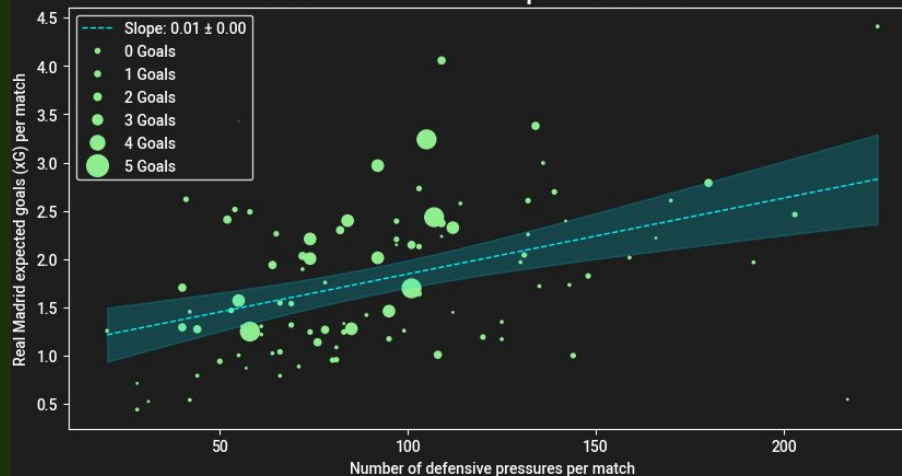
Number of defensive pressures vs goals



Number of defensive pressures vs shots

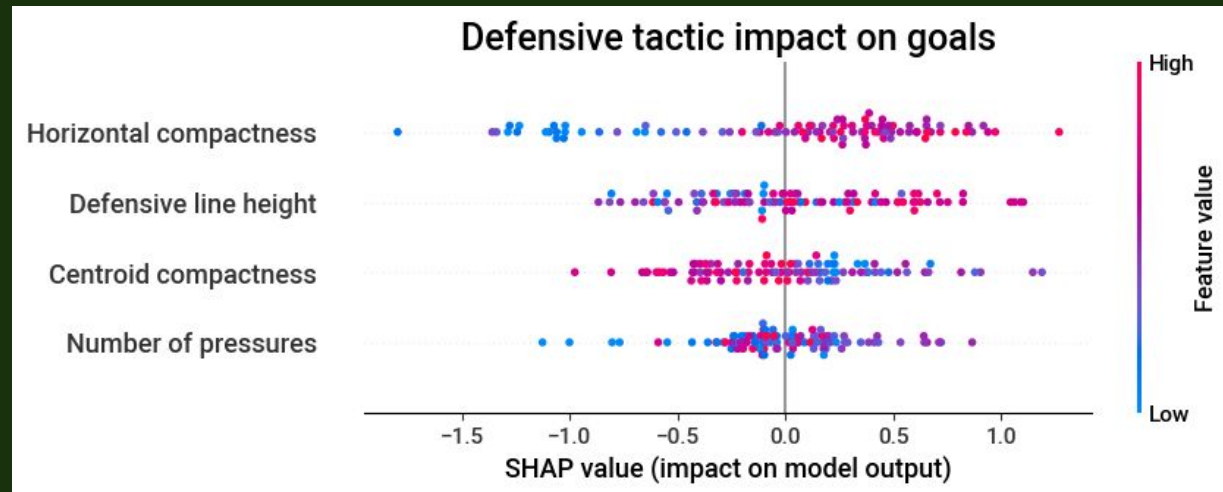
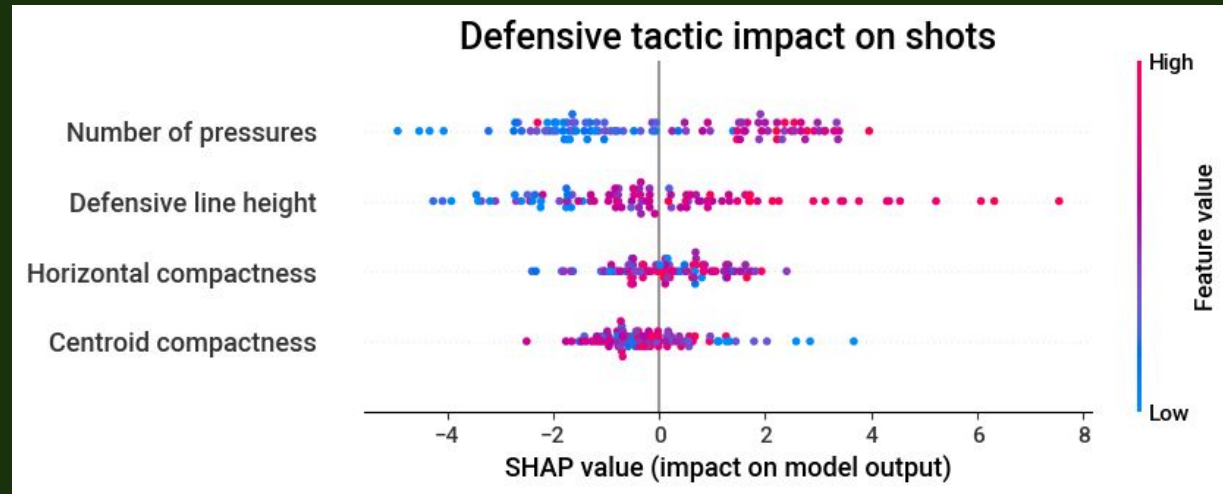


Number of defensive pressures vs xG



Feature Importance

For the 4 major defensive metrics analyzed in this study, a gradient boosted decision tree regression model was trained on all 97 matches to determine which features had a greater impact on shots and goals. The results of the SHAP analysis show that, for shots, the number of pressures had the biggest impact on prediction, with lower numbers of pressures contributing most towards a lower target value (fewer shots). For goals, a lower average horizontal compactness (tighter distance between lines) resulted in the best predictor of fewer goals. Note that the regression model for goals had a negative R-squared value which means that none of these features are a strong indicator for predicting the number of goals in a match.



Examples

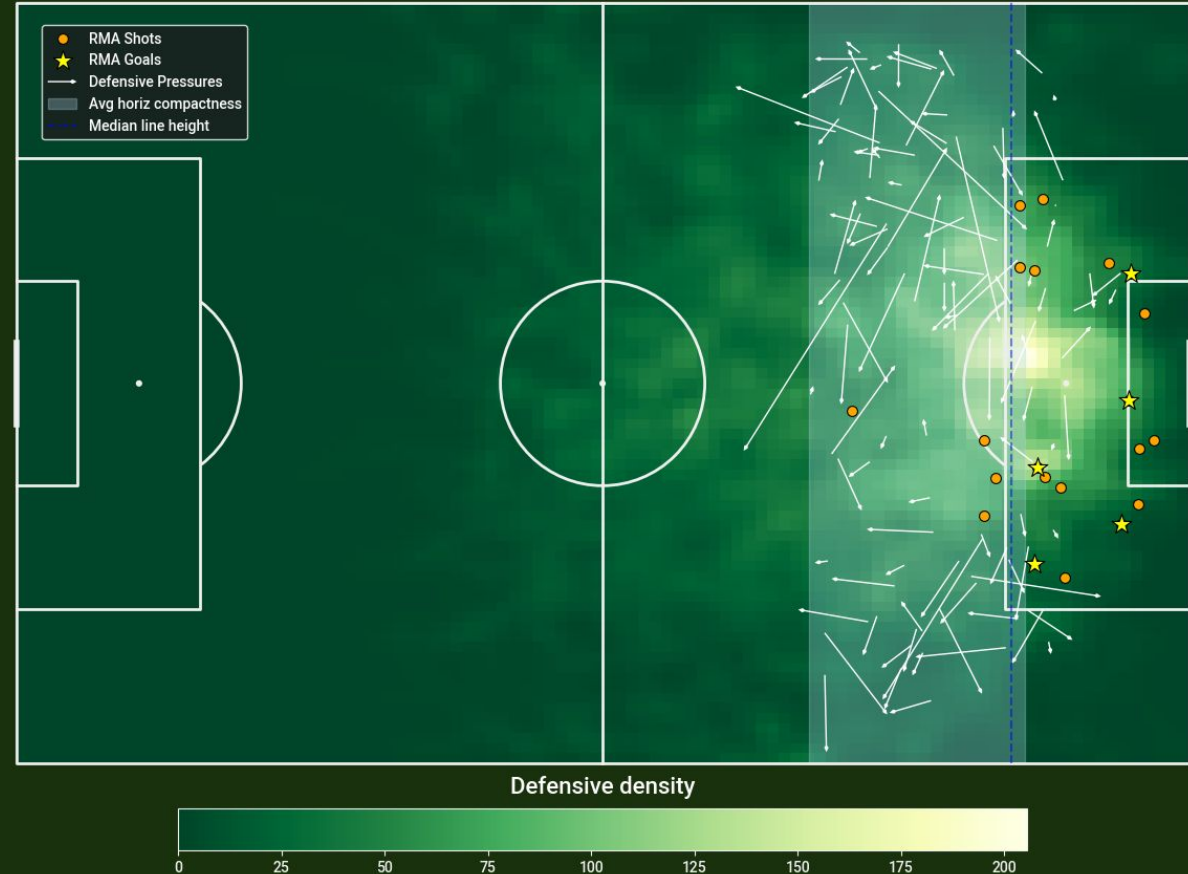
Example 1:

5 RMA goals allowed

This example match from the dataset shows a defensive strategy used by Deportivo Alavés where the defensive line was positioned relatively low, just inside of the penalty area. The defense also committed to over 100 defensive pressure attempts with some breaking outside of the average horizontal compactness zone. Based on the analysis in this report, a high number of shots would be expected, and indeed, Real Madrid was able to take 21 shots and score 5 goals during this match.

Real Madrid 5-0 Alavés, May 14, 2024

Number of shots: 21, Number of goals: 5, Number of pressures: 107, Horizontal compactness: 19.32 meters



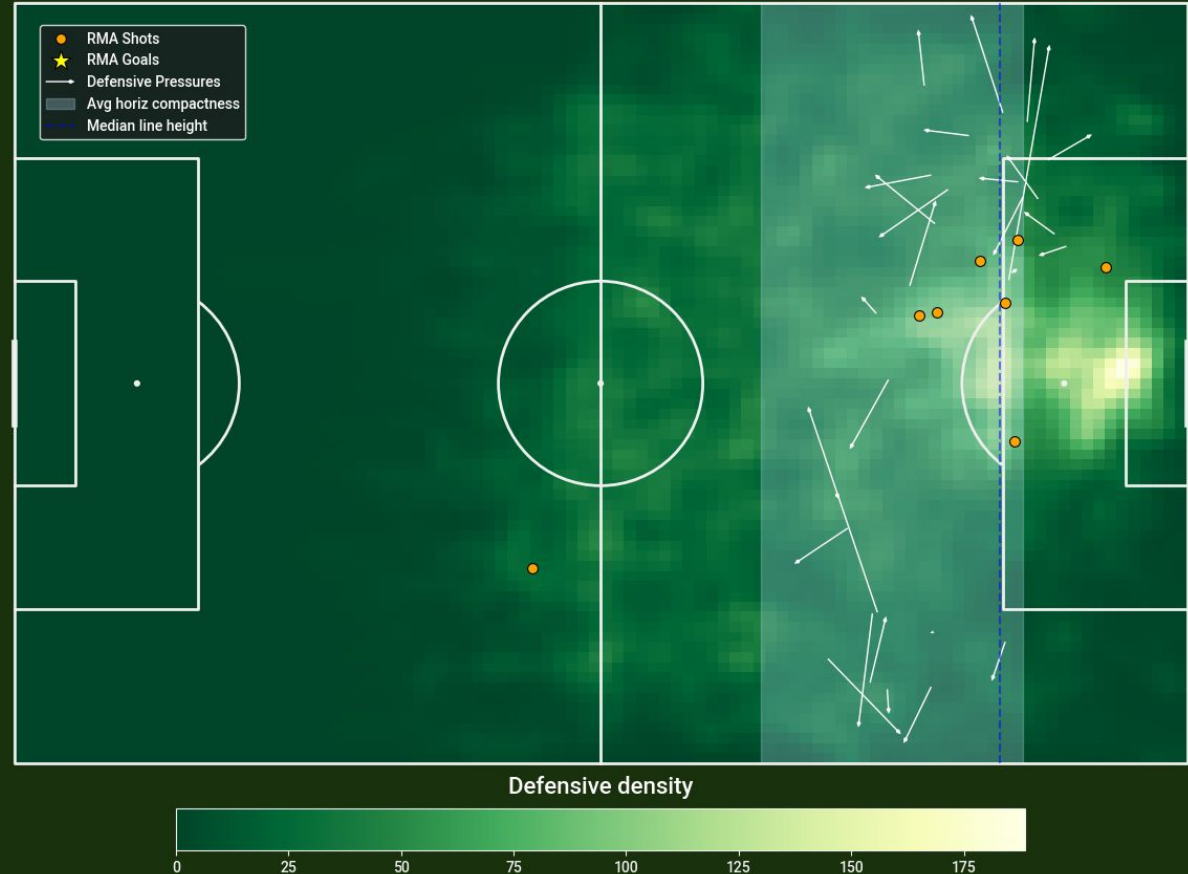
Example 2:

RMA held scoreless

As a counter example, Barcelona kept a slightly higher median defensive line just outside the penalty area. They kept an average horizontal compactness higher up the pitch and committed to a small number of defensive pressures. In fact, Barcelona attempted 80 fewer pressures than Alavés in the previous example. The result is that Real Madrid was limited to only 8 shot attempts and 0 goals.

Real Madrid 0-4 FC Barcelona, October 26, 2024

Number of shots: 8, Number of goals: 0, Number of pressures: 28, Horizontal compactness: 23.44 meters



Appendix

Defensive metrics

For each Real Madrid opponent, 3 defensive metrics were constructed. The goalkeeper was excluded when calculating each of these metrics.

- **Median defensive line height:** the horizontal coordinate positions of the 3 defensive players closest to their own goal were averaged for each frame, then the median was taken for the entire match.
- **Attacks behind defense:** after the median defensive line height was calculated, all shots and goals with coordinates behind this line (closer to defending goal) were collected.
- **Defensive compactness:** 2 different measurements were used for compactness:
 - **Centroid:** the average of all defensive player coordinates was used as a centroid point, then the average distance of each player was calculated. The average of these average distances over the match was used.
 - **Horizontal:** the horizontal (x) separation between the defensive player closest to their own goal and the 8th defensive player closest to their own goal was measured for each frame. The average of these differences over the match was used. Only 8 defensive players were used to avoid outliers of offensive attackers being downfield.
- **Number of pressures:** the number of pressures by the defense in the defensive third when Real Madrid had possession of the ball. Totals were counted from the “pressure” events labeled in the SkillCorner dynamic event data.

Offensive metrics

For each Real Madrid match, 3 offensive metrics were used.

- **Number of shots:** shot totals were aggregated directly from SkillCorner dynamic event data where the “end_type” column value was “shot”.
- **Number of goals:** goal totals were aggregated directly from SkillCorner dynamic event data where the “game_interruption_after” column value was “goal_for”.
- **Expected goals (xG):** expected goals were calculated using all 98 games of the dataset using a logistic regression model fit to the distance of each shot to the goal and angle of each shot to the goal as a features to predict if each shot resulted in a goal.

Data

SkillCorner tracking and dynamic event data is used for this analysis. The dataset includes 97 of Real Madrid's matches during both the 2023/24 and 2024/25 seasons, including both La Liga league matches and Champions League matches.

Scope

Match scenarios analyzed were limited to possessions where the ball was in Real Madrid's attacking half. Averages and medians for each metric over the entire match were used as comparison metrics.

Code location

Notebook code to generate all metrics and graphs located at:

https://github.com/MattLowrieDS/soccermaticspro/blob/main/project_3/project3_defense.ipynb

Visualization app

<https://soccermatics-defense-viz-app.streamlit.app/>